Supplementary information

Network analysis of the genetic relationships between twenty psychiatric and substance use disorders

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World Psychiatry

Table 1 Description of registers

The **Register of Total Population (RTB)**, maintained by Statistics Sweden (SCB), contains data on all individuals registered as residents in Sweden. It is based on the Swedish Tax Agency's population registration system and includes demographic, household, and relational information such as personal identity number, age, sex, marital status, country of birth, citizenship, family and household identifiers, and dates of key life events (e.g., migration, marriage, death). RTB serves as the foundation for official population and household statistics and is updated monthly, with annual person-level data available since 1968 and household data since 2011
Statistics Sweden (SCB). (2024).
Registret över totalbefolkningen (RTB) [Register of Total Population]. Retrieved from https://www.scb.se/vara-tjanster/bestall-data-och-statistik/register/registret-over-totalbefolkningen-rtb/

The **Swedish National Prescribed Drug Register** contains data on all dispensed prescription medications in Sweden since July 2005. It is based on information reported by pharmacies to the Swedish eHealth Agency and includes details such as the Anatomical Therapeutic Chemical (ATC) code, dosage, quantity, date of dispensing, prescriber information, and patient identifiers (encrypted for research use). The register covers all outpatient prescriptions nationwide and is updated monthly. It does not include medications administered during inpatient care.

Swedish National Board of Health and Welfare. (2024). *Läkemedelsregistret [National Prescribed Drug Register]*. Retrieved from https://www.socialstyrelsen.se/statistik-och-data/register/alla-register/lakemedelsregistret/

The **Swedish Population and Housing Census** is a register-based census conducted by Statistics Sweden (SCB). It collects data on households, accommodation, and housing conditions for the entire population at a given point in time. The census integrates information from multiple administrative registers, including the population register and the dwelling register, to produce statistics on household composition, housing characteristics, and living conditions. Sweden transitioned to a fully register-based census after 1990, eliminating the need for questionnaires and enabling annual updates through administrative data.
Statistics Sweden (SCB). (2024). *Register-based Population and Housing Census*. Retrieved from https://www.scb.se/en/finding-statistics/statistics-by-subject-area/population-and-living-conditions/population-composition-and-development/register-based-population-and-housing-census

The Longitudinal Integrated Database for Health Insurance and Labor Market Studies (LISA), maintained by Statistics Sweden (SCB), integrates data from labor market, education, and social insurance registers. It includes all individuals aged ≥16 years (≥15 years since 2010) who were registered as residents in Sweden on December 31 each year, starting in 1990. LISA contains annual information on employment, income, education, sick leave, disability pension,

unemployment benefits, social welfare payments, civil status, and migration. The database was created to support research on health and labour market outcomes and is updated annually Statistics Sweden (SCB). (2019). *LISA – Longitudinal integrated database for health insurance and labour market studies*. Retrieved from https://www.scb.se/contentassets/1bfd261141bc4ddb8e28d4a2e260b690/lisa-bakgrundsfakta-1990-2017.pdf

The **Multi-Generation Register** is a comprehensive dataset comprising individuals who have been registered in Sweden since 1961 or who were born in 1932 or later. These individuals are referred to as index persons, and the register includes links between them and their biological parents. Currently, the register contains information on approximately 11 million index persons. The Multi-Generation Register forms part of the broader Total Population Register, with data sourced from the National Tax Board. Each year, a new version of the register is produced, incorporating new index persons born or immigrating during the year. The data from the Multi-Generation Register can be accessed for research and statistical purposes, offering valuable insights into familial and generational connections within Sweden. Further details about the content and data quality are provided in Statistics Sweden, Background Facts, Population and Welfare Statistics 2017:2, Multi-Generation Register 2016.

The National Patient Register (NPR) was established in the 1960s by the National Board of Health and Welfare to collect data on in-patients at public hospitals in Sweden. Initially, the register focused on psychiatric care patients and about 16% of those receiving somatic (general) care, covering only six of Sweden's 26 county councils. In 1984, the Ministry of Health and Welfare, in collaboration with the Federation of County Councils, decided to make participation in the NPR mandatory for all county councils. By 1987, the register expanded to include all inpatient care across Sweden. Since 2001, NPR has been further broadened to cover outpatient doctor visits, including day surgery and psychiatric care, from both public and private healthcare providers. For more detailed information, you can visit the official site of the National Board of Health and Welfare: https://www.socialstyrelsen.se/en/statistics-and-data/registers/register-information/the-national-patient-register/

We also utilized data from Primary Care, which is a research dataset containing individual-level information on clinical diagnoses from primary healthcare centers in Sweden. By the end of the follow-up period, this dataset covered almost 100% of the population. Figure 1 below illustrates the percentage of the entire Swedish population residing in counties with access to primary care data over time. For more detailed information on this dataset, refer to the study: Sundquist, J., Ohlsson, H., Sundquist, K., Kendler, K.S. (2017) "Common adult psychiatric disorders in Swedish primary care where most mental health patients are treated", published in BMC Psychiatry (17:235). The full article can be accessed here: https://doi.org/10.1186/s12888-017-1381-4.

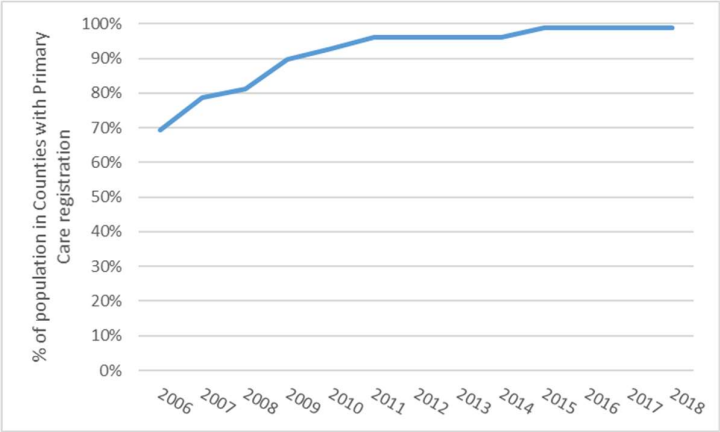

Figure 1 Percentage of the entire Swedish population that resides in counties with primary care data by year

Table 2 Definition of variables

	Registers Used	Definition
Major Depression (MD)	The Swedish Hospital Discharge Register (coverage 1973-2018); Outpatient Care Register (national coverage 2001-2018); Primary Care Data (Partly coverage from 1999-2018)	ICD-8: 296.2, 298.0, 300.4; ICD-9: 296.2, 296.4, 298.0, 300.4; ICD-10: F32, F33.
Drug Use Disorder (DUD)	The Swedish Hospital Discharge Register (coverage 1973-2018); Outpatient Care Register (national coverage 2001-2018); Primary Care Data (Partly coverage from 1999-2018); the Swedish Drug Register (2005-2018); the Swedish Mortality Register, and the Swedish Criminal Register (1973-2018) and the Swedish Suspicion Register (1998-2018)	In the Swedish medical and mortality registries by ICD codes (ICD8: Drug dependence (304); ICD9: Drug psychoses (292) and Drug dependence (304); ICD10: Mental and behavioral disorders due to psychoactive substance use (F10-F19), except those due to alcohol (F10) or tobacco (F17)); in the Suspicion Register by codes 3070, 5010, 5011, and 5012, that reflect crimes related to DA; and in the Crime Register by references to laws covering narcotics (law 1968:64, paragraph 1, point 6) and drug-related driving offences (law 1951:649, paragraph 4, subsection 2 and paragraph 4A, subsection 2). DA was identified in individuals (excluding those suffering from cancer) in the Prescribed Drug Register who had retrieved (in average) more than four defined daily doses a day for 12 months from either of Hypnotics and Sedatives (Anatomical Therapeutic Chemical (ATC) Classification System N05C and N05BA) or Opioids (ATC: N02A).
Alcohol Use Disorder (AUD)	The Swedish Hospital Discharge Register (coverage 1973-2018); Outpatient Care Register (national coverage 2001-2018); Primary Care Data (Partly coverage from 1999-2018); the Swedish Drug Register (2005-2018); the Swedish Mortality Register, and the Swedish Criminal Register (1973-2018) and the Swedish Suspicion Register (1998-2018)	In the Swedish medical and mortality registries by ICD codes: ICD9: V79B, 305A, 357F, 571A-D, 425F, 535D, 291, 303, 980; ICD 10: E244, G312, G621, G721, I426, K292, K70, K852, K860, O354, T51, F10); in the Crime Register by codes 3005, 3201, which reflect crimes related to alcohol abuse; in the Suspicion Register by codes 0004, 0005 (Only those individuals with at least two alcohol-related crimes or suspicion of crimes from both Crime Register and Suspicion Register were included); in the Prescribed Drug Register by the drugs disulfiram (Anatomical Therapeutic Chemical (ATC) Classification System N07BB01), acamprosate (N07BB03), and naltrexone (N07BB04).
Obsessive-Compulsive Disorder (OCD)	The Swedish Hospital Discharge Register (coverage 1973-2018); Outpatient Care Register (national coverage 2001-2018); Primary Care Data (Partly coverage from 1999-2018)	ICD-9: 300D; ICD-10: F42
Attention-Deficit/Hyperactivity Disorder (ADHD)	The Swedish Hospital Discharge Register (coverage 1973-2018); Outpatient Care Register (national coverage 2001-2018); Primary Care	ICD-9: 314; ICD-10: F90

	Data (Partly coverage from 1999-2018)	
Post-Traumatic Stress Disorder (PTSD)	The Swedish Hospital Discharge Register (coverage 1973-2018); Outpatient Care Register (national coverage 2001-2018); Primary Care Data (Partly coverage from 1999-2018)	ICD-10: F431, F430, F620
Phobic Anxiety Disorders (PAD)	The Swedish Hospital Discharge Register (coverage 1973-2018); Outpatient Care Register (national coverage 2001-2018); Primary Care Data (Partly coverage from 1999-2018)	ICD-10: F40
Panic Disorder (PD)	The Swedish Hospital Discharge Register (coverage 1973-2018); Outpatient Care Register (national coverage 2001-2018); Primary Care Data (Partly coverage from 1999-2018)	ICD-10: F410
Generalized Anxiety Disorder (GAD)	The Swedish Hospital Discharge Register (coverage 1973-2018); Outpatient Care Register (national coverage 2001-2018); Primary Care Data (Partly coverage from 1999-2018)	ICD-10: F411
Autism Spectrum Disorder (ASD)	The Swedish Hospital Discharge Register (coverage 1973-2018); Outpatient Care Register (national coverage 2001-2018); Primary Care Data (Partly coverage from 1999-2018)	ICD-10: F840, F841, F845, F849
Bipolar Disorder (BD)	The Swedish Hospital Discharge Register (coverage 1973-2018); Outpatient Care Register (national coverage 2001-2018); Primary Care Data (Partly coverage from 1999-2018).	ICD-8: 296.1, 296.3, 296.8, 296.9, 298.1; ICD-9: 296A, 296C, 296D, 296E, 296W, 298B; ICD-10: F30, F31
Schizophrenia (SZ)	The Swedish Hospital Discharge Register (coverage 1973-2018); Outpatient Care Register (national coverage 2001-2018); Primary Care Data (Partly coverage from 1999-2018)	ICD-8: 295.1, 295.2, 2953, 295.9, 295.6; ICD-9: 295B, 295C, 295D, 295G, 295X; ICD-10: F200, F201, F202, F203, F205, F209
Bulimia Nervosa (BN)	The Swedish Hospital Discharge Register (coverage 1973-2018); Outpatient Care Register (national coverage 2001-2018); Primary Care	ICD-10: F502, F503

	Data(Partly coverage from 1999-2018)	
Anorexia Nervosa (AN)	The Swedish Hospital Discharge Register (coverage 1973-2018); Outpatient Care Register (national coverage 2001-2018); Primary Care Data (Partly coverage from 1999-2018)	ICD-9: 307B; ICD-10: F500
Schizoaffective Disorder (SAD)	The Swedish Hospital Discharge Register (coverage 1973-2018); Outpatient Care Register (national coverage 2001-2018); Primary Care Data (Partly coverage from 1999-2018)	ICD-9: 295H; ICD-10:F25
Acute Psychosis (AP)	The Swedish Hospital Discharge Register (coverage 1973-2018); Outpatient Care Register (national coverage 2001-2018); Primary Care Data (Partly coverage from 1999-2018)	ICD-9: 298E, 298W; ICD-10:F23
Psychosis Not Otherwise Specified (PNOS)	The Swedish Hospital Discharge Register (coverage 1973-2018); Outpatient Care Register (national coverage 2001-2018); Primary Care Data (Partly coverage from 1999-2018)	ICD-9: 298X; ICD-10:F28, F29
Delusional Disorder (DD)	The Swedish Hospital Discharge Register (coverage 1973-2018); Outpatient Care Register (national coverage 2001-2018); Primary Care Data (Partly coverage from 1999-2018)	ICD-9: 297A, 297B, 297C, 297W, 297X; ICD-10:F22
Suicide Attempt (SA)	The Swedish Hospital Discharge Register (coverage 1973-2018); Outpatient Care Register (national coverage 2001-2018); Primary Care Data (Partly coverage from 1999-2018)	ICD-8 codes E950-E959, E980-987 ICD-9 codes E950-E959, E980-987 ICD-10 codes X60-X84, Y10-Y34
Suicide Death (SD)	The Swedish Mortality Register (coverage 1973-2018)	ICD-8 codes E950-E959, E980-987 ICD-9 codes E950-E959, E980-987 ICD-10 codes X60-X84, Y10-Y34
Borderline Personality Disorder (BPD)	The Swedish Hospital Discharge Register (coverage 1973-2018); Outpatient Care Register (national coverage 2001-2018); Primary Care Data (Partly coverage from 1999-2018)	ICD-10 codes F603

Low Educational Attainment (LEA)	LISA	Highest achieved education measured in 1-7 levels that in turn were translated into number of years of education and then standardized with mean 0 and SD 1 by gender and year of birth. 1-Pre-high school (7 years) 2-High School (9 years) 3-Upper Secondary School (11 years) 4-Upper Secondary School (12 years) 5-Post-secondary education (14 years) 6-Post-secondary education (17 years) 7- PhD education (21 years)
Intellectual Disability (ID)	The Swedish Hospital Discharge Register (coverage 1973-2018); Outpatient Care Register (national coverage 2001-2018); Primary Care Data (Partly coverage from 1999-2018)	ICD-8 codes 310, 311, 312, 314, 315 ICD-9 codes 315, 316, 317, 318, 319 ICD-10 codes F71, F72, F73, F78, F79

Table 3 Calculation of the Familial Genetic Risk Score (FGRS)

The dataset for the calculations includes: Column1 = Identification number of the proband (Born 1932-1995) Column2 = Identification number of the relative (1st to 5th degree relatives) Column3 = Proportion of shared additive genetic effects (0.03125 to 0.50) with the proband Column4 = Year of Birth of relative Column5 = Sex of relative Column6 = Age at registration for trait Column7 = Age at end of follow-up (2018-12-31 or age at death, or age at emigration whichever came first)
Step 1: Using all unique relatives with a registration for the disorder, we non-parametrically estimated the distribution of Age at first registration. The empirical distribution is used to obtain weights for relatives without a registration for the disorder, in order to account for the proportion of the time-at-risk period they had completed at the end of follow-up. For example, for relatives at age x at end of follow-up, the weight corresponds to the proportion of relatives registered for the trait that had been registration at age x. For relatives born prior to 1958 we subtracted age at the end of follow-up with the following formula: 1958 - Year of birth of relative. This modification was done in order to control for registration effects (i.e, most registers in Sweden start in 1973 suggesting that relatives from early birth cohorts do not have the possibility to be registered at younger ages). Note that all relatives with the disorder are weighted one.
Step 2: Transform the binary variable (trait yes/no) into a z-score based on the threshold for each trait. The underlying liability of the individual is not assessable. Instead we estimated the mean of the underlying liability to obtain sex and birth decade specific Z-scores for relatives with the trait registration and relatives without the trait. We generate n random numbers from a N(0, 1) distribution and estimate the mean for relatives registered with the disorder (i.e., mean of the observations above the threshold) and for relatives without a registration (i.e., mean of all observation below the threshold). The thresholds are calculated for each decade of birth and sex.
Step 3: Correct for cohabitation effects. To estimate the cohabitation effect (i.e. "shared environment"), we created a database with all individuals in the Swedish population born in Sweden 1955-1990. We also included the number of years, during ages 0-15, that individuals resided in the same household as their biological father. We thereby were able to define two kinds of families i) "not-lived-with" father families (offspring never resided for more than 1 year in the same household or in the same community as their biological father); ii) "lived-with" father (offspring resided a minimum of 13 year in the same household as their biological father. We performed a logistic regression model with the binary trait in offspring as outcome and the binary trait in father, type of father, and their interaction as predictors. We used the interaction term as the difference of effect between genes only and genes + environment. The same approach was performed for half-siblings where we compared those who were reared together versus reared apart
Step 4: Calculate the product for each relative using the four components: i) Z-score (reflecting sex and year of birth adjusted rates) ii) Weight (reflecting the proportion of risk period they had completed) iii) Cohabitation effects iv) Proportion of shared genetic effects (0.03125 – 0.5) with the proband
Step 5: Average the product calculated in step 4 across all relatives to a proband
Step 6: Correct for the number of relatives. We multiplied the results from step 5 with a shrinkage factor. Shrinkage factor (SF): B/(B+A/C). It produces more shrinkage if B and C are small and A is large. (A) the variance of the z-score of the disorder across all relatives, (B) the variance in the mean z-score across all probands, (C) the weighted number of relatives for each proband (sum of Column 3 across each proband).
Step 7: Correct for difference by year of birth and county differences. There are 21 counties in Sweden. For each proband we used the county they had resided in during the maximum number of years (measured from 1969 and onwards) We standardized the risk score by year of birth and county of the proband into a z-score with mean 0 and SD 1.

Table 4 Parameter estimates and CIs for all the edges of the network of the primary model

Node: ADHD

Node	Neighbor	Estimate
ADHD	ASD	0.26 (0.25; 0.26)
ADHD	DUD	0.16 (0.16; 0.17)
ADHD	MD	0.09 (0.09; 0.09)
ADHD	CD	0.08 (0.08; 0.09)
ADHD	AUD	0.06 (0.06; 0.07)
ADHD	PTSD	0.06 (0.06; 0.06)
ADHD	PAD	0.05 (0.05; 0.05)
ADHD	ND	0.05 (0.05; 0.05)
ADHD	BD	0.04 (0.04; 0.04)
ADHD	PD	0.03 (0.03; 0.03)
ADHD	OCD	0.02 (0.02; 0.02)
ADHD	GAD	0.01 (0.01; 0.02)
ADHD	SAD	-0.02 (-0.02; -0.02)
ADHD	SZ	-0.02 (-0.03; -0.02)

Node: AN

Node	Neighbor	Estimate
AN	BN	0.08 (0.08; 0.09)
AN	OCD	0.04 (0.04; 0.04)
AN	MD	0.02 (0.02; 0.03)
AN	PTSD	0.01 (0.01; 0.01)

Node: AP

Node	Neighbor	Estimate
AP	PNOS	0.20 (0.20; 0.21)
AP	DD	0.11 (0.11; 0.12)
AP	SAD	0.08 (0.07; 0.09)
AP	BD	0.07 (0.07; 0.08)
AP	SZ	0.06 (0.06; 0.07)
AP	DUD	0.02 (0.02; 0.03)
AP	ASD	0.02 (0.01; 0.02)
AP	PTSD	0.01 (0.01; 0.01)
AP	MD	0.01 (0.01; 0.01)

Node: ASD

Node	Neighbor	Estimate
ASD	ADHD	0.26 (0.25; 0.26)
ASD	SZ	0.09 (0.09; 0.10)
ASD	OCD	0.05 (0.05; 0.05)
ASD	PNOS	0.05 (0.04; 0.05)
ASD	BD	0.04 (0.04; 0.05)
ASD	MD	0.04 (0.04; 0.04)
ASD	PAD	0.03 (0.03; 0.04)
ASD	SAD	0.03 (0.03; 0.04)
ASD	CD	0.03 (0.03; 0.03)
ASD	AP	0.02 (0.01; 0.02)

Node: AUD

Node	Neighbor	Estimate
AUD	DUD	0.35 (0.34; 0.35)
AUD	ND	0.14 (0.14; 0.14)
AUD	MD	0.07 (0.07; 0.08)
AUD	ADHD	0.06 (0.06; 0.07)
AUD	PTSD	0.05 (0.04; 0.05)
AUD	PD	0.04 (0.04; 0.04)
AUD	BD	0.02 (0.02; 0.03)
AUD	PAD	0.02 (0.02; 0.02)
AUD	PNOS	0.02 (0.01; 0.02)
AUD	SZ	0.02 (0.01; 0.02)
AUD	GAD	0.01 (0.01; 0.02)
AUD	CD	0.01 (0.01; 0.01)
AUD	DD	0.01 (0.01; 0.01)

Node: BD

Node	Neighbor	Estimate
BD	MD	0.14 (0.14; 0.15)
BD	SAD	0.14 (0.13; 0.14)
BD	PNOS	0.08 (0.08; 0.08)
BD	AP	0.07 (0.07; 0.08)
BD	DUD	0.05 (0.05; 0.05)
BD	ASD	0.04 (0.04; 0.05)
BD	ADHD	0.04 (0.04; 0.04)
BD	AUD	0.02 (0.02; 0.03)
BD	PTSD	0.02 (0.02; 0.02)
BD	GAD	0.02 (0.02; 0.02)
BD	OCD	0.01 (0.01; 0.02)
BD	BN	0.01 (0.01; 0.01)

Node: BN

Node	Neighbor	Estimate
BN	AN	0.08 (0.08; 0.09)
BN	MD	0.02 (0.02; 0.02)
BN	OCD	0.01 (0.01; 0.01)
BN	BD	0.01 (0.01; 0.01)
BN	GAD	0.01 (0.01; 0.01)

Node: CD

Node	Neighbor	Estimate
CD	ADHD	0.08 (0.08; 0.09)
CD	ASD	0.03 (0.03; 0.03)
CD	DUD	0.03 (0.02; 0.03)
CD	PTSD	0.01 (0.01; 0.02)
CD	ND	0.01 (0.01; 0.01)
CD	AUD	0.01 (0.01; 0.01)

Node: DD

Node	Neighbor	Estimate
DD	SZ	0.14 (0.14; 0.15)
DD	PNOS	0.14 (0.14; 0.15)
DD	AP	0.11 (0.11; 0.12)
DD	SAD	0.03 (0.02; 0.03)
DD	MD	0.02 (0.02; 0.02)
DD	OCD	0.01 (0.01; 0.02)
DD	DUD	0.01 (0.01; 0.01)
DD	AUD	0.01 (0.01; 0.01)

Node: DUD

Node	Neighbor	Estimate
DUD	AUD	0.35 (0.34; 0.35)
DUD	ADHD	0.16 (0.16; 0.17)
DUD	ND	0.08 (0.08; 0.08)
DUD	PTSD	0.07 (0.07; 0.07)
DUD	MD	0.07 (0.06; 0.07)
DUD	PD	0.05 (0.05; 0.05)
DUD	BD	0.05 (0.05; 0.05)
DUD	GAD	0.05 (0.05; 0.05)
DUD	PNOS	0.04 (0.04; 0.05)
DUD	PAD	0.03 (0.03; 0.03)
DUD	CD	0.03 (0.02; 0.03)
DUD	AP	0.02 (0.02; 0.03)
DUD	SZ	0.02 (0.02; 0.03)
DUD	DD	0.01 (0.01; 0.01)

Node: GAD

Node	Neighbor	Estimate
GAD	PD	0.15 (0.15; 0.16)
GAD	MD	0.15 (0.14; 0.15)
GAD	PAD	0.10 (0.09; 0.10)
GAD	OCD	0.07 (0.07; 0.07)
GAD	DUD	0.05 (0.05; 0.05)
GAD	PTSD	0.03 (0.02; 0.03)
GAD	BD	0.02 (0.02; 0.02)
GAD	ADHD	0.01 (0.01; 0.02)
GAD	AUD	0.01 (0.01; 0.02)
GAD	BN	0.01 (0.01; 0.01)

Node: MD

Node	Neighbor	Estimate
MD	PTSD	0.17 (0.17; 0.18)
MD	GAD	0.15 (0.14; 0.15)
MD	BD	0.14 (0.14; 0.15)
MD	PD	0.13 (0.12; 0.13)
MD	ADHD	0.09 (0.09; 0.09)
MD	PAD	0.08 (0.07; 0.08)
MD	AUD	0.07 (0.07; 0.08)
MD	DUD	0.07 (0.06; 0.07)
MD	OCD	0.06 (0.06; 0.07)
MD	ND	0.05 (0.05; 0.05)
MD	ASD	0.04 (0.04; 0.04)
MD	AN	0.02 (0.02; 0.03)
MD	DD	0.02 (0.02; 0.02)
MD	BN	0.02 (0.02; 0.02)
MD	AP	0.01 (0.01; 0.01)

Node: ND

Node	Neighbor	Estimate
ND	AUD	0.14 (0.14; 0.14)
ND	DUD	0.08 (0.08; 0.08)
ND	ADHD	0.05 (0.05; 0.05)
ND	MD	0.05 (0.05; 0.05)
ND	PD	0.04 (0.03; 0.04)
ND	PTSD	0.03 (0.03; 0.03)
ND	CD	0.01 (0.01; 0.01)

Node: OCD

Node	Neighbor	Estimate
OCD	GAD	0.07 (0.07; 0.07)
OCD	MD	0.06 (0.06; 0.07)
OCD	PAD	0.06 (0.06; 0.06)
OCD	ASD	0.05 (0.05; 0.05)
OCD	AN	0.04 (0.04; 0.04)
OCD	PD	0.03 (0.03; 0.04)
OCD	ADHD	0.02 (0.02; 0.02)
OCD	BD	0.01 (0.01; 0.02)
OCD	DD	0.01 (0.01; 0.02)
OCD	PNOS	0.01 (0.01; 0.01)
OCD	BN	0.01 (0.01; 0.01)

Node: PAD

Node	Neighbor	Estimate
PAD	PD	0.13 (0.13; 0.13)
PAD	GAD	0.10 (0.09; 0.10)
PAD	MD	0.08 (0.07; 0.08)
PAD	OCD	0.06 (0.06; 0.06)
PAD	ADHD	0.05 (0.05; 0.05)
PAD	ASD	0.03 (0.03; 0.04)
PAD	DUD	0.03 (0.03; 0.03)
PAD	AUD	0.02 (0.02; 0.02)
PAD	PTSD	0.01 (0.01; 0.02)

Node: PD

Node	Neighbor	Estimate
PD	GAD	0.15 (0.15; 0.16)
PD	PAD	0.13 (0.13; 0.13)
PD	MD	0.13 (0.12; 0.13)
PD	PTSD	0.06 (0.05; 0.06)
PD	DUD	0.05 (0.05; 0.05)
PD	AUD	0.04 (0.04; 0.04)
PD	ND	0.04 (0.03; 0.04)
PD	OCD	0.03 (0.03; 0.04)
PD	ADHD	0.03 (0.03; 0.03)

Node: PNOS

Node	Neighbor	Estimate
PNOS	SZ	0.21 (0.21; 0.22)
PNOS	AP	0.20 (0.20; 0.21)
PNOS	DD	0.14 (0.14; 0.15)
PNOS	SAD	0.12 (0.11; 0.13)
PNOS	BD	0.08 (0.08; 0.08)
PNOS	ASD	0.05 (0.04; 0.05)
PNOS	DUD	0.04 (0.04; 0.05)
PNOS	AUD	0.02 (0.01; 0.02)
PNOS	OCD	0.01 (0.01; 0.01)

Node: PTSD

Node	Neighbor	Estimate
PTSD	MD	0.17 (0.17; 0.18)
PTSD	DUD	0.07 (0.07; 0.07)
PTSD	ADHD	0.06 (0.06; 0.06)
PTSD	PD	0.06 (0.05; 0.06)
PTSD	AUD	0.05 (0.04; 0.05)
PTSD	ND	0.03 (0.03; 0.03)
PTSD	GAD	0.03 (0.02; 0.03)
PTSD	BD	0.02 (0.02; 0.02)
PTSD	PAD	0.01 (0.01; 0.02)
PTSD	CD	0.01 (0.01; 0.02)
PTSD	AP	0.01 (0.01; 0.01)
PTSD	AN	0.01 (0.01; 0.01)

Node: SAD

Node	Neighbor	Estimate
SAD	SZ	0.21 (0.20; 0.22)
SAD	BD	0.14 (0.13; 0.14)
SAD	PNOS	0.12 (0.11; 0.13)
SAD	AP	0.08 (0.07; 0.09)
SAD	ASD	0.03 (0.03; 0.04)
SAD	DD	0.03 (0.02; 0.03)
SAD	ADHD	-0.02 (-0.02; -0.02)

Node: SZ

Node	Neighbor	Estimate
SZ	PNOS	0.21 (0.21; 0.22)
SZ	SAD	0.21 (0.20; 0.22)
SZ	DD	0.14 (0.14; 0.15)
SZ	ASD	0.09 (0.09; 0.10)
SZ	AP	0.06 (0.06; 0.07)
SZ	DUD	0.02 (0.02; 0.03)
SZ	AUD	0.02 (0.01; 0.02)
SZ	ADHD	-0.02 (-0.03; -0.02)

Table 5 Parameter estimates and CIs for all the edges of the network of the expanded model

Node: ADHD

Node	Neighbor	Estimate
ADHD	ASD	0.24 (0.24; 0.25)
ADHD	DUD	0.14 (0.14; 0.14)
ADHD	LEA	0.11 (0.11; 0.11)
ADHD	MD	0.09 (0.08; 0.09)
ADHD	CD	0.08 (0.07; 0.08)
ADHD	ID	0.06 (0.06; 0.06)
ADHD	BPD	0.05 (0.05; 0.06)
ADHD	PTSD	0.05 (0.05; 0.05)
ADHD	PAD	0.05 (0.04; 0.05)
ADHD	SA	0.04 (0.04; 0.04)
ADHD	AUD	0.04 (0.04; 0.04)
ADHD	BD	0.04 (0.04; 0.04)
ADHD	ND	0.03 (0.02; 0.03)
ADHD	OCD	0.02 (0.02; 0.02)
ADHD	PD	0.02 (0.02; 0.02)
ADHD	GAD	0.01 (0.01; 0.02)
ADHD	SAD	-0.02 (-0.02; -0.02)
ADHD	SZ	-0.02 (-0.03; -0.02)

Node: AN

Node	Neighbor	Estimate
AN	BN	0.08 (0.08; 0.09)
AN	OCD	0.04 (0.04; 0.04)
AN	BPD	0.03 (0.03; 0.04)
AN	MD	0.02 (0.02; 0.02)
AN	SA	0.02 (0.01; 0.02)
AN	LEA	-0.02 (-0.02; -0.02)

Node: AP

Node	Neighbor	Estimate
AP	PNOS	0.20 (0.20; 0.21)
AP	DD	0.11 (0.11; 0.12)
AP	SAD	0.08 (0.07; 0.09)
AP	BD	0.07 (0.07; 0.07)
AP	SZ	0.06 (0.06; 0.07)
AP	DUD	0.02 (0.02; 0.02)
AP	ASD	0.02 (0.01; 0.02)
AP	PTSD	0.01 (0.01; 0.01)
AP	MD	0.01 (0.01; 0.01)

Node: ASD

Node	Neighbor	Estimate
ASD	ADHD	0.24 (0.24; 0.25)
ASD	ID	0.11 (0.11; 0.12)
ASD	SZ	0.09 (0.08; 0.09)
ASD	OCD	0.05 (0.05; 0.05)
ASD	PNOS	0.04 (0.04; 0.05)
ASD	BD	0.04 (0.04; 0.04)
ASD	MD	0.04 (0.03; 0.04)
ASD	SAD	0.03 (0.03; 0.04)
ASD	PAD	0.03 (0.03; 0.04)
ASD	CD	0.03 (0.02; 0.03)
ASD	AP	0.02 (0.01; 0.02)
ASD	SA	0.01 (0.01; 0.01)
ASD	AUD	-0.01 (-0.01; -0.01)

Node: AUD

Node	Neighbor	Estimate
AUD	DUD	0.29 (0.29; 0.29)
AUD	SA	0.17 (0.17; 0.17)
AUD	LEA	0.11 (0.11; 0.11)
AUD	ND	0.11 (0.11; 0.11)
AUD	SD	0.08 (0.08; 0.09)
AUD	MD	0.06 (0.06; 0.06)
AUD	ADHD	0.04 (0.04; 0.04)
AUD	PD	0.03 (0.03; 0.03)
AUD	PTSD	0.03 (0.03; 0.03)
AUD	PAD	0.02 (0.02; 0.02)
AUD	BD	0.01 (0.01; 0.02)
AUD	GAD	0.01 (0.01; 0.02)
AUD	PNOS	0.01 (0.01; 0.02)
AUD	BPD	0.01 (0.01; 0.01)
AUD	SZ	0.01 (0.01; 0.01)
AUD	ASD	-0.01 (-0.01; -0.01)

Node: BD

Node	Neighbor	Estimate
BD	MD	0.14 (0.13; 0.14)
BD	SAD	0.13 (0.13; 0.14)
BD	PNOS	0.08 (0.07; 0.08)
BD	BPD	0.07 (0.07; 0.08)
BD	AP	0.07 (0.07; 0.07)
BD	SA	0.05 (0.05; 0.05)
BD	ASD	0.04 (0.04; 0.04)
BD	ADHD	0.04 (0.04; 0.04)
BD	DUD	0.04 (0.03; 0.04)
BD	SD	0.03 (0.03; 0.04)
BD	GAD	0.02 (0.01; 0.02)
BD	AUD	0.01 (0.01; 0.02)
BD	PTSD	0.01 (0.01; 0.02)
BD	OCD	0.01 (0.01; 0.02)
BD	LEA	-0.05 (-0.05; -0.04)

Node: BN

Node	Neighbor	Estimate
BN	AN	0.08 (0.08; 0.09)
BN	BPD	0.04 (0.03; 0.04)
BN	MD	0.02 (0.01; 0.02)
BN	OCD	0.01 (0.01; 0.01)
BN	LEA	-0.02 (-0.02; -0.02)

Node: BPD

Node	Neighbor	Estimate
BPD	SA	0.09 (0.08; 0.09)
BPD	BD	0.07 (0.07; 0.08)
BPD	DUD	0.06 (0.06; 0.06)
BPD	ADHD	0.05 (0.05; 0.06)
BPD	PTSD	0.05 (0.04; 0.05)
BPD	BN	0.04 (0.03; 0.04)
BPD	AN	0.03 (0.03; 0.04)
BPD	MD	0.02 (0.02; 0.03)
BPD	PAD	0.02 (0.02; 0.03)
BPD	OCD	0.02 (0.02; 0.02)
BPD	GAD	0.02 (0.02; 0.02)
BPD	PNOS	0.01 (0.01; 0.02)
BPD	PD	0.01 (0.01; 0.02)
BPD	AUD	0.01 (0.01; 0.01)

Node: CD

Node	Neighbor	Estimate
CD	ADHD	0.08 (0.07; 0.08)
CD	ASD	0.03 (0.02; 0.03)
CD	ID	0.03 (0.02; 0.03)
CD	LEA	0.02 (0.02; 0.03)
CD	DUD	0.02 (0.02; 0.02)
CD	SA	0.02 (0.01; 0.02)
CD	PTSD	0.01 (0.01; 0.01)

Node: DD

Node	Neighbor	Estimate
DD	SZ	0.14 (0.14; 0.15)
DD	PNOS	0.14 (0.14; 0.15)
DD	AP	0.11 (0.11; 0.12)
DD	SAD	0.03 (0.02; 0.03)
DD	MD	0.02 (0.02; 0.02)
DD	OCD	0.01 (0.01; 0.02)
DD	DUD	0.01 (0.01; 0.01)

Node: DUD

Node	Neighbor	Estimate
DUD	AUD	0.29 (0.29; 0.29)
DUD	SA	0.15 (0.15; 0.16)
DUD	ADHD	0.14 (0.14; 0.14)
DUD	LEA	0.07 (0.07; 0.08)
DUD	BPD	0.06 (0.06; 0.06)
DUD	ND	0.06 (0.06; 0.06)
DUD	PTSD	0.05 (0.05; 0.06)
DUD	MD	0.05 (0.05; 0.05)
DUD	SD	0.05 (0.04; 0.05)
DUD	GAD	0.05 (0.04; 0.05)
DUD	PD	0.04 (0.04; 0.04)
DUD	PNOS	0.04 (0.03; 0.04)
DUD	BD	0.04 (0.03; 0.04)
DUD	PAD	0.03 (0.02; 0.03)
DUD	AP	0.02 (0.02; 0.02)
DUD	CD	0.02 (0.02; 0.02)
DUD	SZ	0.02 (0.01; 0.02)
DUD	DD	0.01 (0.01; 0.01)

Node: GAD

Node	Neighbor	Estimate
GAD	PD	0.15 (0.15; 0.16)
GAD	MD	0.15 (0.14; 0.15)
GAD	PAD	0.10 (0.09; 0.10)
GAD	OCD	0.07 (0.07; 0.07)
GAD	DUD	0.05 (0.04; 0.05)
GAD	PTSD	0.03 (0.02; 0.03)
GAD	BPD	0.02 (0.02; 0.02)
GAD	BD	0.02 (0.01; 0.02)
GAD	ADHD	0.01 (0.01; 0.02)
GAD	AUD	0.01 (0.01; 0.02)

Node: ID

Node	Neighbor	Estimate
ID	ASD	0.11 (0.11; 0.12)
ID	LEA	0.08 (0.08; 0.08)
ID	ADHD	0.06 (0.06; 0.06)
ID	CD	0.03 (0.02; 0.03)
ID	PNOS	0.02 (0.02; 0.03)
ID	ND	0.02 (0.02; 0.02)
ID	SA	0.01 (0.01; 0.01)
ID	PTSD	0.01 (0.01; 0.01)

Node: LEA

Node	Neighbor	Estimate
LEA	ND	0.15 (0.15; 0.15)
LEA	AUD	0.11 (0.11; 0.11)
LEA	ADHD	0.11 (0.11; 0.11)
LEA	ID	0.08 (0.08; 0.08)
LEA	DUD	0.07 (0.07; 0.08)
LEA	PD	0.05 (0.05; 0.06)
LEA	SA	0.05 (0.05; 0.05)
LEA	PTSD	0.02 (0.02; 0.03)
LEA	CD	0.02 (0.02; 0.03)
LEA	PAD	0.02 (0.02; 0.02)
LEA	AN	-0.02 (-0.02; -0.02)
LEA	BN	-0.02 (-0.02; -0.02)
LEA	OCD	-0.02 (-0.03; -0.02)
LEA	MD	-0.03 (-0.03; -0.03)
LEA	BD	-0.05 (-0.05; -0.04)

Node: MD

Node	Neighbor	Estimate
MD	PTSD	0.17 (0.16; 0.17)
MD	GAD	0.15 (0.14; 0.15)
MD	BD	0.14 (0.13; 0.14)
MD	PD	0.13 (0.12; 0.13)
MD	ADHD	0.09 (0.08; 0.09)
MD	PAD	0.08 (0.07; 0.08)
MD	SA	0.06 (0.06; 0.07)
MD	OCD	0.06 (0.06; 0.06)
MD	AUD	0.06 (0.06; 0.06)
MD	DUD	0.05 (0.05; 0.05)
MD	ND	0.05 (0.05; 0.05)
MD	ASD	0.04 (0.03; 0.04)
MD	SD	0.03 (0.02; 0.03)
MD	BPD	0.02 (0.02; 0.03)
MD	AN	0.02 (0.02; 0.02)
MD	DD	0.02 (0.02; 0.02)
MD	BN	0.02 (0.01; 0.02)
MD	AP	0.01 (0.01; 0.01)
MD	SZ	-0.01 (-0.01; -0.01)
MD	LEA	-0.03 (-0.03; -0.03)

Node: ND

Node	Neighbor	Estimate
ND	LEA	0.15 (0.15; 0.15)
ND	AUD	0.11 (0.11; 0.11)
ND	DUD	0.06 (0.06; 0.06)
ND	MD	0.05 (0.05; 0.05)
ND	SA	0.03 (0.03; 0.03)
ND	PD	0.03 (0.03; 0.03)
ND	ADHD	0.03 (0.02; 0.03)
ND	PTSD	0.02 (0.02; 0.02)
ND	ID	0.02 (0.02; 0.02)

Node: OCD

Node	Neighbor	Estimate
OCD	GAD	0.07 (0.07; 0.07)
OCD	MD	0.06 (0.06; 0.06)
OCD	PAD	0.06 (0.06; 0.06)
OCD	ASD	0.05 (0.05; 0.05)
OCD	AN	0.04 (0.04; 0.04)
OCD	PD	0.03 (0.03; 0.04)
OCD	BPD	0.02 (0.02; 0.02)
OCD	ADHD	0.02 (0.02; 0.02)
OCD	DD	0.01 (0.01; 0.02)
OCD	BD	0.01 (0.01; 0.02)
OCD	PNOS	0.01 (0.01; 0.01)
OCD	BN	0.01 (0.01; 0.01)
OCD	LEA	-0.02 (-0.03; -0.02)

Node: PAD

Node	Neighbor	Estimate
PAD	PD	0.13 (0.12; 0.13)
PAD	GAD	0.10 (0.09; 0.10)
PAD	MD	0.08 (0.07; 0.08)
PAD	OCD	0.06 (0.06; 0.06)
PAD	ADHD	0.05 (0.04; 0.05)
PAD	ASD	0.03 (0.03; 0.04)
PAD	DUD	0.03 (0.02; 0.03)
PAD	BPD	0.02 (0.02; 0.03)
PAD	LEA	0.02 (0.02; 0.02)
PAD	AUD	0.02 (0.02; 0.02)
PAD	PTSD	0.01 (0.01; 0.01)
PAD	SA	0.01 (0.01; 0.01)

Node: PD

Node	Neighbor	Estimate
PD	GAD	0.15 (0.15; 0.16)
PD	PAD	0.13 (0.12; 0.13)
PD	MD	0.13 (0.12; 0.13)
PD	LEA	0.05 (0.05; 0.06)
PD	PTSD	0.05 (0.05; 0.06)
PD	DUD	0.04 (0.04; 0.04)
PD	OCD	0.03 (0.03; 0.04)
PD	AUD	0.03 (0.03; 0.03)
PD	ND	0.03 (0.03; 0.03)
PD	ADHD	0.02 (0.02; 0.02)
PD	BPD	0.01 (0.01; 0.02)

Node: PNOS

Node	Neighbor	Estimate
PNOS	SZ	0.21 (0.21; 0.22)
PNOS	AP	0.20 (0.20; 0.21)
PNOS	DD	0.14 (0.14; 0.15)
PNOS	SAD	0.12 (0.11; 0.13)
PNOS	BD	0.08 (0.07; 0.08)
PNOS	ASD	0.04 (0.04; 0.05)
PNOS	DUD	0.04 (0.03; 0.04)
PNOS	ID	0.02 (0.02; 0.03)
PNOS	SA	0.02 (0.01; 0.02)
PNOS	BPD	0.01 (0.01; 0.02)
PNOS	AUD	0.01 (0.01; 0.02)
PNOS	OCD	0.01 (0.01; 0.01)

Node: PTSD

Node	Neighbor	Estimate
PTSD	MD	0.17 (0.16; 0.17)
PTSD	SA	0.06 (0.06; 0.06)
PTSD	PD	0.05 (0.05; 0.06)
PTSD	DUD	0.05 (0.05; 0.06)
PTSD	ADHD	0.05 (0.05; 0.05)
PTSD	BPD	0.05 (0.04; 0.05)
PTSD	AUD	0.03 (0.03; 0.03)
PTSD	GAD	0.03 (0.02; 0.03)
PTSD	LEA	0.02 (0.02; 0.03)
PTSD	ND	0.02 (0.02; 0.02)
PTSD	SD	0.02 (0.01; 0.02)
PTSD	BD	0.01 (0.01; 0.02)
PTSD	PAD	0.01 (0.01; 0.01)
PTSD	CD	0.01 (0.01; 0.01)
PTSD	ID	0.01 (0.01; 0.01)
PTSD	AP	0.01 (0.01; 0.01)

Node: SA

Node	Neighbor	Estimate
SA	AUD	0.17 (0.17; 0.17)
SA	DUD	0.15 (0.15; 0.16)
SA	BPD	0.09 (0.08; 0.09)
SA	MD	0.06 (0.06; 0.07)
SA	PTSD	0.06 (0.06; 0.06)
SA	SD	0.06 (0.05; 0.06)
SA	BD	0.05 (0.05; 0.05)
SA	LEA	0.05 (0.05; 0.05)
SA	ADHD	0.04 (0.04; 0.04)
SA	ND	0.03 (0.03; 0.03)
SA	SZ	0.02 (0.02; 0.03)
SA	PNOS	0.02 (0.01; 0.02)
SA	AN	0.02 (0.01; 0.02)
SA	CD	0.02 (0.01; 0.02)
SA	ID	0.01 (0.01; 0.01)
SA	PAD	0.01 (0.01; 0.01)
SA	ASD	0.01 (0.01; 0.01)
SA	SAD	0.01 (0.01; 0.01)

Node: SAD

Node	Neighbor	Estimate
SAD	SZ	0.21 (0.20; 0.22)
SAD	BD	0.13 (0.13; 0.14)
SAD	PNOS	0.12 (0.11; 0.13)
SAD	AP	0.08 (0.07; 0.09)
SAD	ASD	0.03 (0.03; 0.04)
SAD	DD	0.03 (0.02; 0.03)
SAD	SA	0.01 (0.01; 0.01)
SAD	ADHD	-0.02 (-0.02; -0.02)

Node: SD

Node	Neighbor	Estimate
SD	AUD	0.08 (0.08; 0.09)
SD	SA	0.06 (0.05; 0.06)
SD	DUD	0.05 (0.04; 0.05)
SD	BD	0.03 (0.03; 0.04)
SD	MD	0.03 (0.02; 0.03)
SD	SZ	0.02 (0.02; 0.03)
SD	PTSD	0.02 (0.01; 0.02)

Node: SZ

Node	Neighbor	Estimate
SZ	PNOS	0.21 (0.21; 0.22)
SZ	SAD	0.21 (0.20; 0.22)
SZ	DD	0.14 (0.14; 0.15)
SZ	ASD	0.09 (0.08; 0.09)
SZ	AP	0.06 (0.06; 0.07)
SZ	SA	0.02 (0.02; 0.03)
SZ	SD	0.02 (0.02; 0.03)
SZ	DUD	0.02 (0.01; 0.02)
SZ	AUD	0.01 (0.01; 0.01)
SZ	MD	-0.01 (-0.01; -0.01)
SZ	ADHD	-0.02 (-0.03; -0.02)

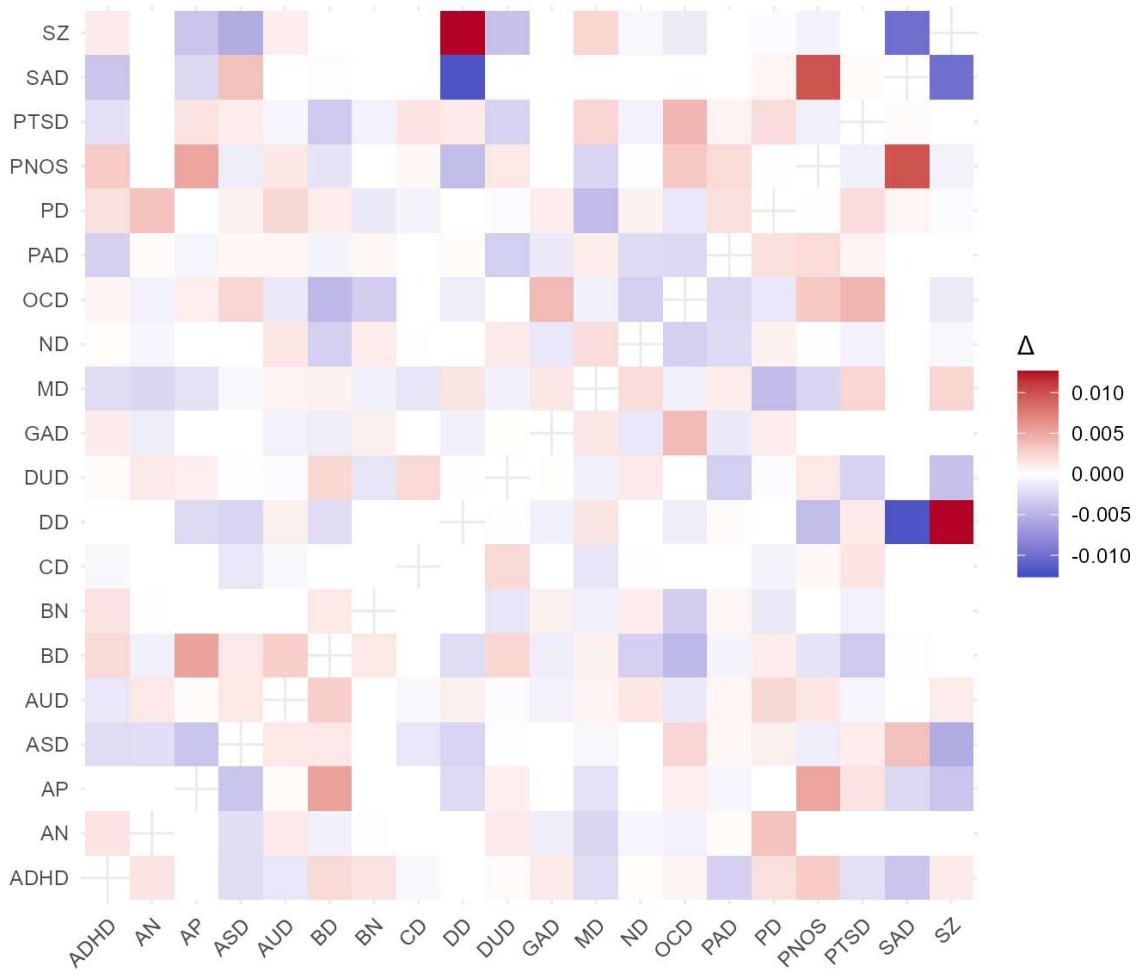

Figure 2 Heat map comparing edge estimates across our split-halves of the primary model (not significant differences, p=0.08)

Tables 6 and 7

Results for the comparison of our edge estimates in primary model in birth cohorts from 1960-1980 and 1981-2000

Var1	Var2	Difference: Young-Old	p
DUD	AUD	0.052	0.006
ASD	ADHD	-0.039	0.006
ASD	SZ	0.029	0.006
AP	PNOS	0.023	0.006
PTSD	MD	0.022	0.006
BN	AN	0.018	0.006
ND	ADHD	0.015	0.006
CD	ADHD	-0.014	0.006
ADHD	DUD	0.014	0.006
DUD	MD	-0.012	0.006
ADHD	AUD	0.012	0.006
ASD	BD	0.012	0.006
ND	DUD	0.011	0.006
CD	ND	0.01	0.006
ADHD	BD	0.009	0.006
AN	ADHD	0.009	0.006
ND	AUD	0.009	0.006

Node	Difference: Young-Old	P_value
DUD	0.085	0.001
AUD	0.082	0.001
PNOS	0.048	0.001
AN	0.040	0.001
ND	0.040	0.001
BN	0.036	0.001
SZ	0.032	0.001
GAD	0.031	0.001
PD	0.026	0.001
PTSD	0.025	0.001
PAD	0.024	0.001
MD	0.023	0.001
AP	0.022	0.001
ADHD	0.015	0.014

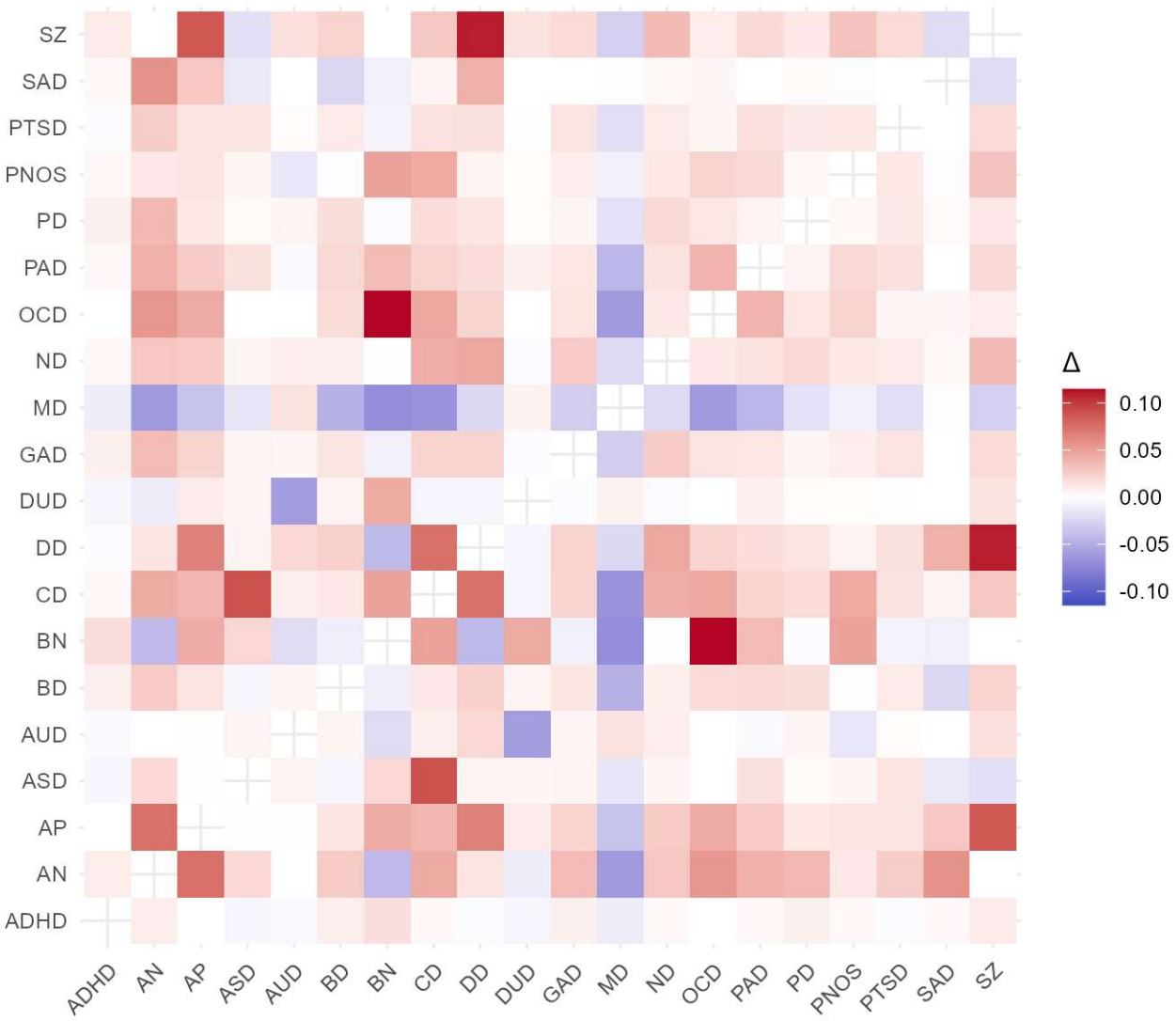

Figure 3 Heat map comparing edge estimates from our original analysis compared with those generated with the use of a Yeo-Johnson transformation (correlation of edge weights: +0.88)

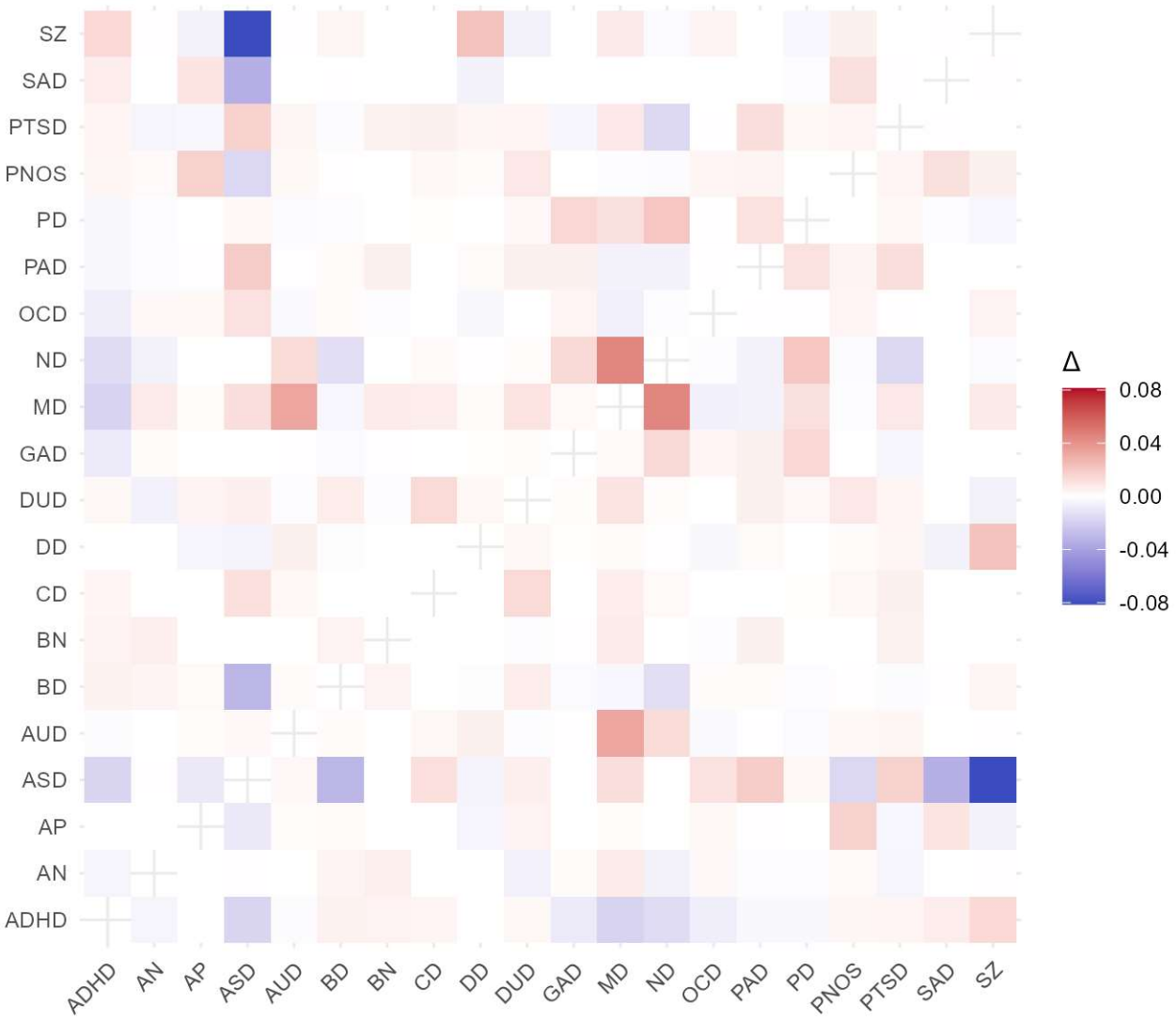

Figure 4 Heat map comparing edge estimates from our original analysis with the standard FGRS compared with those generated using the PA-FGRS of Morten et al (correlation of edge weights: +0.98)